Owner Training Agenda

Danfoss VLT HVAC Drive Operation & Maintenance

Location: 1490 METRA 47TH STREET DIESEL SHOP – 147 West 47th Street, Chicago, IL
Equipment Covered:

- Three (3) **Danfoss VLT HVAC 5 HP, 460V, EMB2, CB DRIVES**
- Two (2) **Danfoss VLT HVAC 25 HP, 460V/60/3 CB DRIVES**

Training Focus: Owner operation, routine maintenance, basic troubleshooting, and safe handling of Danfoss VLT HVAC drives

1. Introduction & System Overview

- Introduction to the Danfoss VLT HVAC drive platform
- Review of the five installed VFDs and their associated HVAC equipment
- Purpose and benefits of VFDs in HVAC applications
- How the VFD interacts with the motor, HVAC equipment, and building controls/BAS
- Identification of each drive and its associated equipment

2. VFD Safety & Basic Electrical Awareness

- Electrical hazards associated with VFD equipment
- Proper shutdown and isolation procedures
- Importance of allowing the drive's DC bus capacitors to discharge before servicing
- Understanding drive status indicators and warning signs
- What owner personnel should and should not attempt to service
- When to contact a qualified technician

3. Danfoss VLT HVAC Drive Controls & Display

- Overview of the local control panel/keypad
- Navigating the drive display and menus
- Understanding:
    - Start/Stop status
    - Hand/Auto operation

    - Frequency/speed
    - Motor current
    - Output voltage
    - Fault and warning indications
- Understanding the difference between **Hand**, **Off**, and **Auto** modes
- Returning the drive to normal automatic operation after testing

4. Normal Operation

- Starting and stopping the HVAC equipment
- Normal operating indications
- Monitoring drive speed/frequency
- Understanding how the building automation system commands the drive
- Recognizing normal versus abnormal operating conditions
- Demonstration of local Hand operation, where appropriate
- Proper procedure for returning equipment to BAS/Auto control

5. Basic Troubleshooting

- Review of common Danfoss VLT warnings and alarms
- How to identify and document a fault code
- Reading the drive's fault/alarm history
- Basic checks before resetting a fault
- Demonstration of an appropriate fault reset
- Conditions under which a fault **should not** simply be reset
- Common causes of drive trips, including:
    - Motor overload
    - Overcurrent
    - Overvoltage/undervoltage
    - Overtemperature

    - Loss of input power
    - Motor or wiring issues
    - External/BAS control issues
- What information to provide Bornquist when requesting service

6. Building Automation System (BAS) Interface

- How the BAS communicates with the VFD
- Review of the drive's automatic operating sequence
- Understanding speed/frequency commands
- Start/stop commands
- Alarm/fault feedback to BAS
- Recognizing whether an issue originates at the drive, motor, or BAS
- Importance of avoiding unauthorized parameter changes

7. Routine Maintenance & Inspection

- Recommended visual inspection of the VFDs
- Checking for:
    - Excessive dust/debris
    - Overheating
    - Unusual noise
    - Loose or damaged components
    - Abnormal smells
    - Cooling/ventilation issues
    - Fault or warning indications
- Importance of maintaining adequate airflow around the drives
- Cleaning considerations and what **not** to do when cleaning VFD equipment
- Reviewing drive operating history and fault history as part of maintenance

8. Parameter & Programming Overview

- General overview of Danfoss VLT parameters
- Parameters that are useful for owner/operator monitoring
- Identifying motor nameplate information
- Review of application-specific settings
- Explanation of why critical parameters should not be changed without authorization
- Backup/documentation of important drive settings

9. Hands-On Owner Exercises

Owner personnel will have an opportunity to practice:

- Navigating the drive keypad
- Identifying the current operating mode
- Reading operating values
- Checking frequency/speed
- Viewing active alarms and warnings
- Reviewing fault history
- Performing a controlled fault reset
- Switching between appropriate operating modes
- Returning the drive to normal BAS/Auto operation
- Identifying the appropriate information to record when reporting a problem

10. Maintenance & Service Best Practices

- Recommended owner inspection frequency
- Keeping drive enclosures and surrounding areas clean
- Maintaining proper ventilation
- Monitoring recurring alarms and faults
- Maintaining records of drive faults and service
- Avoiding unnecessary parameter changes
- Recognizing conditions requiring professional service

11. Q&A & Review

- Review of key operating procedures
- Review of common alarms and troubleshooting steps
- Owner questions regarding the five installed drives
- Review of emergency/service procedures
- Confirmation that owner personnel can:
    - Identify the drives
    - Determine operating status
    - Read basic operating information
    - Identify a fault
    - Safely document the fault
    - Perform basic permitted troubleshooting
    - Return the system to normal automatic operation

12. Training Completion

- Final questions and answers
- Review of recommended maintenance practices
- Provide owner with applicable Danfoss documentation/manuals
- Document attendees and completion of owner training

Estimated Training Duration: >1 hour

Training Objective:
At the conclusion of the training, owner personnel should have a working understanding of the five Danfoss VLT HVAC drives, including normal operation, keypad navigation, BAS interaction, basic troubleshooting, routine visual maintenance, fault identification, and appropriate escalation of service issues.